MADT 7202 MID-TERM · CHAMPION SYNERGY NETWORK · HIGH-ELO SOLO QUEUE, SEA + KR

Ban the pair, not the champion? — a LoL synergy network for tournament draft prep (Patch 26.20)

1 The decision

Which 2–3 champion pairs a Thai amateur/collegiate team should ban or practise at its next tournament. Owner: **head coach / draft analyst**. If top-synergy pairs were just high-win-rate champions, win-rate bans would do.

2 Why a network

A champion's win rate is one row in a table. Synergy — winning more together than the two win rates predict — belongs to a pair, so it lives only on the edges of a champion network.

3 Data

Source / method	Riot API (league-v4 → match-v5), Challenger/GM/Master solo queue, patch 26.20, 6 Oct–15 Oct 2026: SEA (Thai queue since the 2025 TH2 merge) and KR
Permission	Riot developer API terms; Leaguepedia CC BY-SA; no names, Riot IDs or PUUIDs kept
Volume	12,598 API calls; 10,997 matches (5,997 SEA + 5,000 KR), remakes out; 173 champions
Sampling bias	Solo queue ≠ 5-stack, KR ≠ Thai: SEA and KR checked apart and against 182 pro games

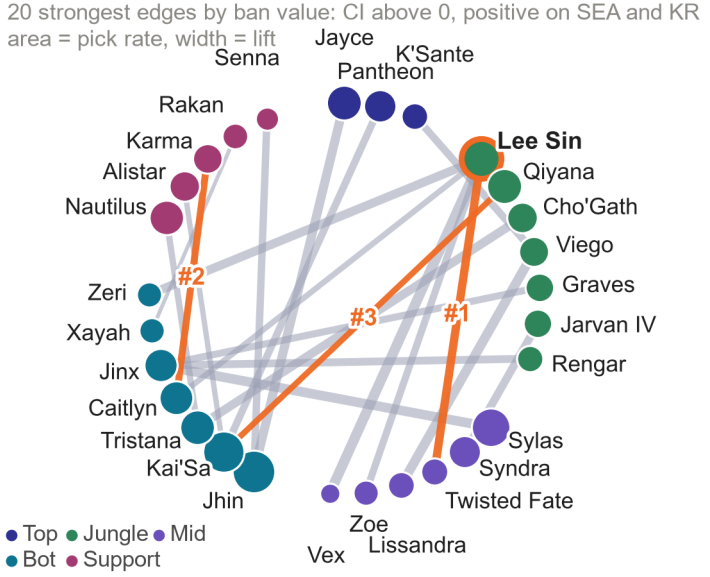
4 Graph construction

Node / edge	Champion; edge = teammates in one game; 8,787 → 638 edges with ≥ 30 games
Weight	Lift = pair WR − σ(logit WR(a) + logit WR(b)), pp. Raw co-pick puts meta pairs first (Lulu + Yunara: n=393, -0.4 pp); 0 of its top 5 are in the lift top 5
E-role vs E-all	E-role keeps bot lane and jungle–top/mid/bot pairs; E-all (all 10) shares 0/3 of the top 3 and puts 2 non-interacting pairs in its top 5
Shrinkage / score	WR + Beta(10,10), pair prior 20 games; ban value = lift × pair games / matches; CI = 200 bootstraps

5 The picture

Data: Riot Games API, SEA + KR, patch 26.20

20 strongest edges by ban value: CI above 0, positive on SEA and KR
area = pick rate, width = lift



Ban list: top ban value, 90% CI above 0

	games	lift
#1 Lee Sin + Twisted Fate (Jg-Mid)	92	+10.0
#2 Caitlyn + Karma (Bot-Sup)	112	+8.0
#3 Kai'Sa + Qiyana (Bot-Jg)	133	+6.5

90% CI: #1 +3.3 to +15.6, #2 +1.9 to +14.1, #3 +0.6 to +12.2.
Pro games won on 26.20: #1 not played, #2 2 of 12, #3 2 of 2.
CI above 0: 43 of 638 pairs (random: mean 36, max 46).
Hub Lee Sin: 7 of 30 solid edges (random max 6); betweenness 0.33; own win rate rank 90 of 173.

Top 5 by raw co-pick count

	games	lift
1. Lulu + Yunara	393	-0.4
2. Kai'Sa + Nautilus	271	-2.7
3. Aphelios + Thresh	246	+3.7
4. Ezreal + Karma	234	+1.8
5. Lucian + Milio	232	+1.7

Top 5 by synergy lift (≥ 30 games)

	games	lift
1. Lee Sin + Vex	37	+12.8
2. Fiddlesticks + Jhin	31	+12.6
3. Gragas + Lee Sin	32	+12.6
4. Jinx + Sylas	49	+11.9
5. Lissandra + Viego	56	+11.8

6 Findings

- **Pair synergy is weak.** 43 of 638 pairs have a CI above 0, but so do 36 on average (max 46) when winners are re-drawn at random. Top ban value: **Lee Sin + Twisted Fate** (Jg-Mid) +10.0 pp (CI +3.3 to +15.6, n=92).
- **The hub is real:** Lee Sin holds 7 of the 30 edges that pass on both servers (random max 6), betweenness 0.33, yet ranks 90th by own win rate; no ban-list champion is top-10 by win rate, so a win-rate table misses them. The most-picked champion, Jhin, is in a top lift pair.
- **Robustness:** the 3 pairs stay top-5 at 20/30/50 games (3/3/3 of 3), without shrinkage (3/3), E-all 2/3; lift is positive in all four SEA/KR and tier halves (3/3). But Caitlyn + Karma won 2 of 12 pro games.

7 Recommended action

- Scrims 19–25 Oct 2026: first-phase ban Lee Sin (cuts 7 synergy edges incl. pair #1); owner: draft analyst.
- Practise Kai'Sa + Qiyana in 10 scrims by 31 Oct 2026, logging win rate with/without it; drop Caitlyn + Karma.
- Re-run on patch day (21 Oct 2026, 26.21); if the Lee Sin hub falls to the random level, ban by win rate.

8 What would change my mind

Solo-queue synergy may not carry into coordinated play. If 10 scrims show opponents win as often with Lee Sin banned, or 26.21 pro games disagree, I would ban by win rate alone.

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